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RESEARCH PROFILE

I study how neural circuits transform noisy, distributed sensory information into a single committed action. My postdoctoral work established two results. First, visual approach detecting neurons integrate cues in the natural temporal order those cues occur during a real approach event, a previously unrecognized strategy I termed *sequenced cue integration*. Second, as co-lead of a collaborative project, I discovered the first neural circuit in any animal that computes the direction of motion of an odor signal. My laboratory will build on these to study multisensory integration, behavioral arbitration under conflicting drives, and bilateral sensory comparison, using *Drosophila* as a system where these questions are quantitatively tractable. I combine quantitative behavior, two-photon imaging, optogenetics, genetic perturbation, and modeling, with a background in experimental and theoretical physics.

TRAINING AND EDUCATION

Yale University, New Haven, CT, USA Postdoctoral Associate in Neuroscience. Mentor: Dr. Damon Clark Visual and olfactory sensory computation in <i>Drosophila</i>	2021 – present
University of Pittsburgh, Pittsburgh, PA, USA Ph.D. in Physics. Advisor: Dr. Hanna Salman Thesis: “The roles of intergeneration inheritance and intrageneration molecular dynamics in shaping living cells.”	2015 – 2021
Indian Institute of Science Education and Research, Bhopal, India B.S.–M.S. in Physics. Advisor: Dr. Dhanvir S. Rana Thesis: “Induction of ferromagnetic ordering in paramagnetic CaRuO ₃ by intrinsic tensile strain.”	2010 – 2015
Marine Biological Laboratory, Woods Hole, MA, USA Methods in Computational Neuroscience (selective summer course)	Summer 2022

GRANTS AND FELLOWSHIPS

K99/R00 Pathway to Independence application — National Institutes of Health (NINDS) “Neural mechanisms underlying survival-driven conflicts in <i>Drosophila</i> .” Impact score 23; Just-in-Time information requested; council review completed (Sep. 2026).	2026
Provost’s Fellowship , University of Pittsburgh	2021
Andrew W. Mellon Predoctoral Fellowship , University of Pittsburgh	2020
Summer Research Fellowship , Indian Academy of Sciences	2013
INSPIRE Fellowship , Department of Science and Technology, India	2010 – 2015

AWARDS AND HONORS

Winner, Three-Minute Thesis (3MT) Competition, University of Pittsburgh	2019
Travel Grant, International Summer School on Theoretical Biophysics, Cargèse, France	2017

PUBLICATIONS

Thirteen research articles and preprints, five as first or co-first author. Author name in bold; * denotes equal contribution.

First and co-first author publications and preprints

1. Santana, G. *, **Vashistha, H. ***, Emonet, T., & Clark, D. A. (2026). A neural circuit for olfactory motion detection. *Under review at Nature*. *Equal contribution. [bioRxiv](#)
2. **Vashistha, H.**, Matos, N. C., Wu, H., & Clark, D. A. (2026). Dynamic trajectory cues drive sequenced integration in approach detectors. *In revision at PNAS*. [bioRxiv](#)
3. **Vashistha, H.**, & Clark, D. A. (2026). A luminance-induced expansion aftereffect. *Perception* (in press). [link](#)
4. **Vashistha, H.**, Jammal-Touma, J., Singh, K., Rabin, Y., & Salman, H. (2023). Bacterial cell-size changes resulting from altering the relative expression of Min proteins. *Nature Communications*, 14, 5710. [link](#)
5. **Vashistha, H.**, Kohram, M., & Salman, H. (2021). Non-genetic inheritance restraint of cell-to-cell variation. *eLife*, 10, e64779. [link](#) (*eLife* Digest: “How long is a cell’s memory?” [link](#))

Contributing author publications

6. Kohram, M., Sanderson, A., Loui, A., Thompson, P. V., **Vashistha, H.**, Shomar, A., Oltvai, Z. N., & Salman, H. (2025). Non-lethal deleterious mutation-induced allostasis accelerates bacterial aging. *PNAS*, 121(20), e2316271121. [link](#)
7. Knapp, B. D., Willis, L., Gonzalez, C., **Vashistha, H.**, Jammal-Touma, J., Tikhonov, M., Ram, J., Salman, H., Elias, J. E., & Huang, K. C. (2024). Metabolic rearrangement enables adaptation of microbial growth rate to temperature shifts. *Nature Microbiology*, 10, 185–201. [link](#)
8. ElGamel, M., **Vashistha, H.**, Salman, H., & Mugler, A. (2023). Multigenerational memory in cell size homeostasis. *Physical Review E*, 108, L032401. [link](#)
9. Stawsky, A., **Vashistha, H.**, Nechleba, J. T., Salman, H., & Brenner, N. (2022). Multiple timescales in bacterial growth homeostasis. *iScience*, 25, 103678. [link](#)
10. Kohram, M., **Vashistha, H.**, Leibler, S., Xue, B., & Salman, H. (2021). Bacterial growth control mechanisms inferred from multivariate statistical analysis of single-cell measurements. *Current Biology*. [link](#)
11. Rashid, S., Long, Z., Singh, S., Kohram, M., **Vashistha, H.**, Navlakha, S., Salman, H., Oltvai, Z. N., & Bar-Joseph, Z. (2019). Adjustment in tumbling rates improves bacterial chemotaxis on obstacle-laden terrains. *PNAS*, 116(24). [link](#)
12. Susman, L., Kohram, M., **Vashistha, H.**, Nechleba, J. T., Salman, H., & Brenner, N. (2018). Individuality and slow dynamics in bacterial growth homeostasis. *PNAS*, 115(25). [link](#)

Invited commentary

13. **Vashistha, H.**, & Clark, D. A. (2022). Feature maps: how the insect visual system organizes information. *Current Biology* (invited Dispatch). [link](#)

INVITED TALKS AND SEMINARS

Jawaharlal Nehru Centre for Advanced Scientific Research, Bengaluru, India	2025
<i>Dynamic trajectory cues drive sequenced cue integration in approach detectors</i>	
Indian Institute of Technology Madras, Chennai, India	2025
<i>Visual feature integration to detect approach in flies and humans</i>	
Stanford University, Palo Alto, CA, USA	2019
<i>Non-genetic inheritance in isogenic bacterial populations</i>	

SELECTED CONFERENCE PRESENTATIONS

Computational and Systems Neuroscience (COSYNE), Lisbon, Portugal <i>Visual feature integration to detect approach in flies and humans (poster)</i>	2026
Neurobiology of Drosophila, Cold Spring Harbor Laboratory, NY, USA <i>Sequenced cue integration in approach detectors (poster)</i>	2025
Indian Drosophila Research Conference, IISER Mohali, India <i>Sequenced cue integration in approach detectors (poster)</i>	2025
Kavli Institute Symposium, Yale University, USA <i>Multi-cue integration in approach detection (poster)</i>	2024
Marine Biological Laboratory, Woods Hole, MA, USA <i>Building a steering signal in the fly brain (talk)</i>	2022
International Summer School on Theoretical Biophysics, Cargèse, France <i>Individuality and non-genetic inheritance in isogenic bacterial cells (talk)</i>	2017

TECHNICAL EXPERTISE

Neurophysiology and circuit manipulation: two-photon calcium imaging; optogenetic activation and silencing; Gal4/UAS and split-Gal4 genetic targeting; connectome-guided circuit analysis.

Behavior and instrumentation: closed-loop fly-on-a-ball behavioral rigs with panoramic visual displays; optogenetic odor delivery systems with arbitrary spatiotemporal statistics; human and fly psychophysics; microfluidic device design and fabrication; long-term fluorescence microscopy.

Computational: Python and MATLAB; quantitative behavioral analysis; time-series and multivariate statistics; dynamical-systems and phenomenological modeling.

MENTORING

All Yale trainees were co-supervised with Dr. Damon Clark.

Research project supervision

Natalia CB Matos — Graduate student, Yale University 2021 – present
Escape responses to translational motion and the fly's cutting strategy; manuscript in preparation, with N. Matos as first author and myself as second.

Valentina Zhang — Graduate student, Yale University 2026 – present
Binocular visual processing in the fly; guided the design of anaglyph visual stimuli and of protocols for fly behavioral assays.

Krystian Loetscher — Graduate student, Yale University Spring 2022
Developed a new protocol for imaging the neck connective; provided training in dissection and imaging.

Daniel Otchere — Undergraduate, Yale University Summer 2022
Three months full-time; designed and built a rig for measuring threat-evoked behavior in freely moving flies, analyzed with custom-made automated tracking pipelines.

Method and technical training

Kate Maier — Graduate student, Yale University 2023 – present
Two-photon calcium imaging and visual stimulus design.

Dr. Maitri Manjunath — Postdoctoral researcher, Yale University Summer 2025
Programming, quantitative behavioral methods, and two-photon imaging.

Parisa Vaziri — Undergraduate, Yale University 2022 – 2024

Habituation of the fly's escape response to looming stimuli; trained in programming and quantitative behavioral analysis. Now a Ph.D. student at Harvard University.

Dr. Joanna Jammal-Touma — Postdoctoral researcher, University of Pittsburgh Summer 2021
Soft lithography, microfluidic fabrication, and single-cell analysis of bacterial growth and division.

Peyton Thompson — Undergraduate, University of Pittsburgh Summer 2017
PCR, cell growth analysis, and general laboratory methods.

TEACHING

Neuroscience Bootcamp for Ph.D. students Summer 2022 – present
Teaching Assistant, Yale University

Basic Physics for Science and Engineering (PHY 0174) Spring 2016
Teaching Assistant, University of Pittsburgh

Introductory Physics (PHY 101) Summer 2016
Teaching Assistant, University of Pittsburgh

Stars, Galaxies, and the Cosmos (ASTRO 0089) Fall 2015, 2016
Teaching Assistant, University of Pittsburgh

SERVICE AND OUTREACH

Peer review: Co-reviewed a manuscript for *PLOS Biology* with Dr. Damon Clark.

Teacher, IFET organization 2020
Designed and taught online science courses to high school students in rural India during the COVID-19 lockdown.

NSF Outreach Program, Pittsburgh Public School District 2017
Worked with high school teachers and students to develop instructional materials for science education.

REFERENCES

1. **Hanna Salman, Ph.D.**

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214 Old Engineering Hall, 3943 O'Hara Street, Pittsburgh, PA 15260, USA
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Ph.D. advisor, University of Pittsburgh, 2015–2021.

2. **Damon A. Clark, Ph.D.**

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Postdoctoral advisor, Yale University, 2021–present.

3. **Thierry Emonet, Ph.D.**

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Collaborator and co-author on Santana, Vashistha et al. (2026), Yale University.